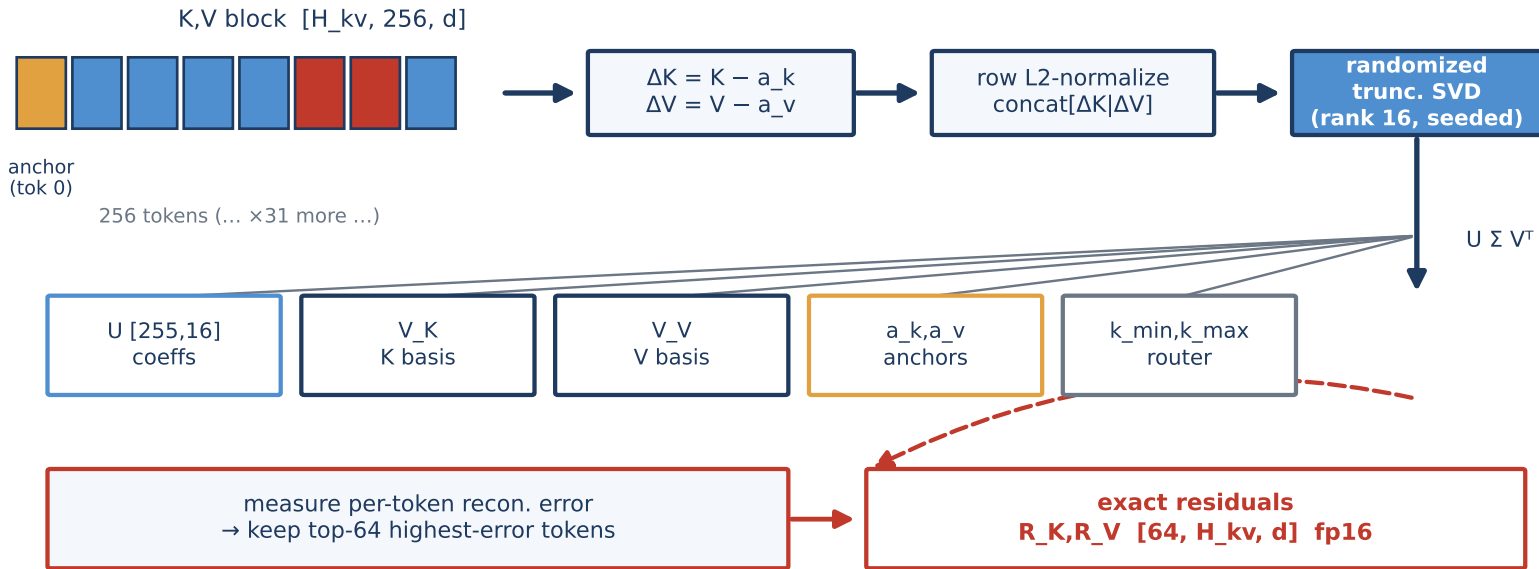


Differential KV Compression — one 256-token block

Anchor + low-rank joint SVD delta (shared structure) + top-error exact residuals (the outliers SVD discards)



stored block ≈ 90 KiB (26 KiB low-rank + 64 KiB residuals) vs 256 KiB dense $\rightarrow \approx 2.85\times$ smaller

low-rank recon. (implicit): $\hat{K}_i \approx a_k + s \cdot (U_i \cdot V_K)$; residuals attended EXACTLY at decode $\rightarrow$ verbatim recall